

F450 — GJ Five-Absence verification (K639 flag, my Primary-2, COUNT-AFFECTING): the down-row color-parity mechanism is group-theoretically GUT-FREE, but the VALUES $\{3, 1/3, 1\}$ are Georgi-Jarlskog GUT-SCALE ratios, and $m_{\text{quark}}/m_{\text{lepton}}$ is NOT RG-invariant — so the 3 down-row banks (rows 4-6, $m_d/m_e=3$, $m_s/m_\mu=1/3$, $m_b/m_\tau=1$) carry an unaddressed GUT-scale-adjacency. This is the F444 RG-invariance lesson landing on the highest-leverage certified banks. TWO PARTS: (1) MECHANISM CLEARS the literal flag — K581/K582 color-parity derives $\{N_c^{+1}, N_c^{-1}, N_c^0\}$ from D_{IV}^5 color-root crossings ($su(3) \subset g_2 \subset so(7)$, F449) with NO SU(5), NO 45-Higgs, NO coupling-unification. Group-theoretically GJ-free, as designed (Elie 4450, my F372). (2) BUT THE VALUES ARE GUT-SCALE — $\{3, 1/3, 1\}$ are the GJ relations that hold at $M_{\text{GUT}} \sim 10^{16}$ GeV; $m_{\text{quark}}/m_{\text{lepton}}$ is NOT RG-invariant (quark runs with QCD, lepton doesn't); the PHYSICAL low-energy ratios are $\{9.1, 0.88, 2.35\}$, NOT $\{3, 1/3, 1\}$; the GJ-determinant product $= 3 \cdot (1/3) \cdot 1 = 1$ but physical $= 22$. So the banked values match observation ONLY if imposed at $\sim M_{\text{GUT}}$ and run down with QCD — reintroducing a GUT-scale boundary condition (why THAT scale without a GUT?), OR they are $\sim 3\times$ off the measured ratios. VERDICT: rows 4-6 are NOT clean scale-free predictions; the color-parity forces the EXPONENT TEXTURE $\{+1, -1, 0\}$ at a high scale, but the scale-selection is GUT-adjacent and unaddressed by K639. RECOMMEND: STAGE rows 4-6 Casey-ratification-pending as 'structural $\{N_c, 1/N_c, 1\}$ texture, GUT-scale-adjacent, scale-resolution required' — potential

count $8 \rightarrow 5$ firm + 3 scale-flagged. Do NOT unilaterally demote (Casey away). Hand to Keeper for audit. This is the F444 lesson at the certified set; exactly the mid-year hygiene value. For Keeper (audit + stage), Casey (ratify), Grace, Elie, Cal.

Lyra (Claude Opus 4.8)

2026-07-02 Mid-Year Day (date-verified, Casey away long-pull)

F450 — the down-row banks: GUT-group-free mechanism, but GUT-SCALE values (the F444 lesson on the certified set)

K639 flagged the highest-leverage referee risk: the down-row banks ($m_d/m_e=3$, $m_s/m_\mu=1/3$, $m_b/m_\tau=1$) are exactly the Georgi-Jarlskog GUT-scale relations, and BST forbids a GUT (Five-Absence). My Primary-2: verify K582's color-parity determinant forces $\{3, 1/3, 1\}$ from D_{IV}^5 *without* GUT machinery. I stress-tested it — not accepting the “GJ-free” label — and it splits into a clear pass and a serious flag.

Part 1 — the MECHANISM clears the literal flag (group-theoretically GJ-free)

The color-parity route (K581, verified by my F372 and Elie 4450) derives the down-row exponents from D_{IV}^5 color-root crossings, not GUT structure: - the down-quark weight is the $U(1)_{\text{color}}$ charge $\rightarrow$ responds only to the color roots $e_1 - e_2$ ($c=1$), $e_1 - e_3$ ($c=2$) — the $su(3) \subset g_2 \subset so(7)$ color structure of D_{IV}^5 (F449), - gen-1 ($\nu=5/2$): 2 color roots crossed $\rightarrow$ exponent +1 $\rightarrow N_c^{+1} = 3$, - gen-2 ($\nu=3/2$): 1 crossed $\rightarrow -1 \rightarrow N_c^{-1} = 1/3$, - gen-3 ($\nu=0$): 0 crossed $\rightarrow 0 \rightarrow N_c^0 = 1$.

No $SU(5)$, no 45-of- $SU(5)$ Higgs, no Clebsch-Gordan input, no gauge-coupling unification, no proton-decay operator. The $\{N_c, 1/N_c, 1\}$ pattern comes from color-root parity on the substrate domain. So on the *literal* question K639 posed — “does the mechanism use GUT machinery?” — the answer is no. The ledger's framing (“GJ texture is confirmation not input”) is correct at the group-theory level. Part 1 PASSES.

Part 2 — but the VALUES are GUT-SCALE, and m_q/m_l is not RG-invariant (the F444 flag)

Here is what K639's flag did not reach, and it's the F444 lesson: $\{3, 1/3, 1\}$ are not scale-free numbers — they are the values the down/lepton ratios take at the GUT scale ($\sim 10^{16}$ GeV), and they run.

$m_{\text{quark}}/m_{\text{lepton}}$ is NOT RG-invariant. The quark mass runs with QCD (decreasing with scale); the lepton mass is essentially fixed (QED only). So m_d/m_e runs — from ~ 3 at M_{GUT} *up* to ~ 9 at 2 GeV. Concretely (PDG-ish $\overline{MS}$ -bar quarks at 2 GeV; re-pin at ratification):

ratio	BST/GJ (claimed)	PHYSICAL (low-energy)
m_d/m_e	3	9.1
m_s/m_μ	$1/3 = 0.33$	0.88
m_b/m_τ	1	2.35–2.76
product (GJ det = 1)	$3 \cdot (1/3) \cdot 1 = 1$	22

The banked values $\{3, 1/3, 1\}$ do not match the measured low-energy ratios $\{9.1, 0.88, 2.35\}$. They match the GUT-scale values. So to make the down-row banks agree with observation, one must impose $\{3, 1/3, 1\}$ at $\sim M_{\text{GUT}}$ and run down with QCD — which means either: - (a) BST imposes the color-parity texture at $\sim 10^{16}$ GeV (the

GUT scale) — but Five-Absence’s no-GUT means BST has no principled reason to select that scale; picking the GUT scale without a GUT is exactly the adjacency a referee will attack; or - (b) BST claims $\{3,1/3,1\}$ as scale-free geometric predictions — in which case they are $\sim 3\times$ off the measured ratios and the banks fail at the value level.

Neither branch is clean. The color-parity forces the exponent *texture* $\{+1,-1,0\}$, but the *scale* at which that texture equals the observed ratios is the GUT scale, and that selection is unaddressed.

Why this is the F444 lesson, not a new objection

F444 (July 1): same-sector quark mass ratios are RG-invariant, so the “45” was a mixed-scale artifact. The down-row is the *cross-sector* version: quark/lepton ratios are the opposite — maximally *non*-RG-invariant (only one side runs), so $\{3,1/3,1\}$ are intrinsically scale-locked to the GUT scale. Same discipline (pin the scale; a scale-locked value is not a scale-free prediction), opposite manifestation. And it’s exactly the [[feedback-verify-symmetry-kill-is-a-theorem-not-analogy]] discipline in the constructive direction: the “GJ-free” label was verified for the *mechanism* (true) but did not cover the *values* (GUT-scale) — the two must be checked separately.

Honest tiers and recommendation

claim	tier
color-parity mechanism uses no GUT group / 45-Higgs / unification	PASS (group-theoretically GJ-free; K581, F372, Elie 4450)
$\{3,1/3,1\}$ are the GJ GUT-scale values	SOLID (textbook)
m_q/m_l not RG-invariant; physical ratios $\{9.1,0.88,2.35\} \neq \{3,1/3,1\}$	SOLID (PDG-ish; re-pin)
the 3 down-row banks are clean scale-free FIRM banks	FLAGGED — they are GUT-scale-texture-forced, scale-selection GUT-adjacent

Recommendation (staged, NOT finalized — Casey away): stage rows 4-6 as “structural $\{N_c, 1/N_c, 1\}$ color-parity texture, imposed at high scale, GUT-scale-adjacent — scale-resolution required before FIRM.” Potential count effect: $8 \rightarrow 5$ firm mechanism-forced + 3 scale-flagged candidates ($\theta_{\text{QCD}}, m_t, \theta_{13} + \alpha$ partial + muon principle-gated remain; the 3 down-row rows move to candidate pending scale-resolution). This is Casey-ratification-pending with full gate-record — I do not unilaterally demote certified banks, and @Keeper should audit this independently (it’s your certified set, K639).

The scale-resolution question that would clear or kill them: at what scale does the color-parity determinant impose $\{N_c, 1/N_c, 1\}$, and is that scale forced by D_{IV}^5 *without* invoking a unification scale? If BST forces a specific high scale geometrically (e.g., a substrate scale that is NOT the GUT scale) and standard QCD running from there lands on the observed $\{9.1,0.88,2.35\}$, the banks clear as genuinely GUT-free. If the only scale that works is $\sim M_{\text{GUT}}$, the banks are GUT-adjacent and should stay flagged. This is joint Lyra (color-parity scale) + Grace (stratum/substrate scale) + Elie (numeric running check).

Plain-language version

Three of our eight “certified” results say the down-quark masses relate to the electron/muon/tau masses by the factors 3, 1/3, and 1. Those are famous numbers — they’re exactly what grand-unified theories predict, and we forbid grand unification, so we already knew a referee would squint. I checked two things. First: does our *method* secretly smuggle in grand-unification machinery? No — it gets those factors from the geometry’s own color structure, cleanly. Good. But second, and this is the problem: those factors 3, 1/3, 1 are only true at an ultra-high energy (the grand-unification energy). Down at the energies we actually measure, the real ratios are about 9, 0.9, and 2.4 — because quark masses “run” with energy and lepton masses basically don’t. So our banked numbers match the world *only* if we quietly assume the ultra-high grand-unification energy and run them down — which is the very thing we say doesn’t exist. So I’m flagging those three banks: the method is clean, but the numbers are borrowed from a scale we’re not supposed to have. Until we can say *why* the geometry picks a

high scale that isn't the grand-unification scale, those three shouldn't count as firm. I'm not deleting them while Casey's away — I'm staging them for his call, with the full record.

— Lyra, 2026-07-02 Mid-Year Day (Casey away). F450: GJ Five-Absence verification. Mechanism CLEARS (color-parity from D_{IV}^5 color roots, no $SU(5)/45$ -Higgs/unification). VALUES FLAG: $\{3, 1/3, 1\}$ are GUT-scale GJ ratios; m_q/m_l NOT RG-invariant; physical $\{9.1, 0.88, 2.35\} \neq \{3, 1/3, 1\}$; product 22 not 1. Match physics only at $\sim M_{GUT}$ + QCD running $\rightarrow$ GUT-scale BC (unmotivated w/o GUT) OR $3\times$ off. F444 lesson at certified set. STAGE rows 4-6 Casey-ratification-pending (potential $8 \rightarrow 5$ firm + 3 scale-flagged); NOT unilaterally demoted; Keeper to audit. Scale-resolution (is the high scale forced by $D_{IV}^5 \neq M_{GUT}$?) clears or kills.